

Modelling & Simulation: Assignment 7

Control of a 3-D robot arm

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Introduction

Now that you've gained some experience with both 2-D robot arms and 3-D rigid bodies, it is time to combine the two: a 3-D robot arm with 3 rotational degrees of freedom.

However, rather than focusing on modelling the dynamics, in this assignment you will focus on 3-D kinematics and Cartesian impedance control of the robot arm. Therefore, you can model the arm using 20-sim's built-in 3D Mechanics Editor. We are also going to ignore the actuator dynamics; your joints may be 'actuated' by MSe elements.

Suggested material:

- Albu-Schäffer et al., "Cartesian Impedance Control Techniques for Torque Controlled Light-Weight Robots", IEEE International Conference on Robotics and Automation (ICRA), 2002;
- Albu-Schäffer et al., "Cartesian Impedance Control of Redundant Robots: Recent Results with the DLR-Light-Weight-Arms", IEEE International Conference on Robotics and Automation (ICRA), 2003;
- Canvas: 3D Rotation 2018 V2 negative angles.pdf.
- Canvas: Representing Attitude: Euler Angles, Unit Quaternions and Rotation Vectors.pdf
- Canvas: geometry_and_screw_theory_for_robotics.pdf

Assignment

Model a 3-D, 3-DoF robot arm with rotational joints using 20-sim's 3D Mechanics Editor. Choose the joint axes such that the end-effector can move in 3-D (i.e. not planar), and add torque inputs on the joints using MSe elements. Using the 3-D kinematics of the robot, implement Cartesian impedance control and demonstrate it in a simulation experiment, where the robot follows a repeated circular trajectory in 3-D with its end-effector.

Some questions you could try to answer in the process:

- Is there a 'best' choice for the joint axes? (e.g. yaw-pitch-roll, yaw-pitch-pitch, etc.?)
- Will you include centrifugal/coriolis terms, or are they negligible at the speeds you are considering?
- Are you going to compensate gravity with your control? Can you show its effect in the tracking of your robot's end-effector?
- What impedance parameters will you choose for your Cartesian impedance control and why?